

CYBER SECURITY INTERNSHIP

Task 1: Understanding Cyber Security Basics & Attack Surface

Name: Jeffrina V

Internship Organization: Elevate Labs

Domain: Cyber Security

Task: Task 1

Date: 15 January 2026

1. What is Cyber Security?

Cyber Security is the practice of protecting systems, networks, and data from digital attacks. These attacks aim to access, change, or destroy sensitive information.

The core principles of cyber security are explained by the CIA Triad.

2. CIA Triad

Confidentiality

Ensures that data is accessed only by authorized users.

Example:

Banking apps use passwords and OTPs to protect account details.

Integrity

Ensures data is accurate and not altered without permission.

Example:

Transaction amount should not change during online payment.

Availability

Ensures systems and data are available when needed.

Example:

Email services should be accessible 24/7 without downtime.

3. Types of Cyber Attackers

Script Kiddies

Beginners who use ready-made hacking tools without deep knowledge.

Insiders

Employees or trusted users who misuse access intentionally or accidentally.

Hacktivists

Attack systems to promote political or social causes.

Nation-State Attackers

Government-sponsored attackers targeting other countries' systems.

4. Attack Surface

An attack surface is any point where an attacker can try to enter or extract data from a system.

Common Attack Surfaces:

- Web Applications
- Mobile Applications
- APIs
- Networks(Wi-Fi, Routers)
- Cloud Infrastructure

5. OWASP Top 10 (Overview)

OWASP Top 10 lists the most critical web application security risks.

Some major vulnerabilities include:

- SQL Injection
- Broken Authentication
- Cross-Site Scripting (XSS)
- Security Misconfiguration

Sensitive Data Exposure

These vulnerabilities are dangerous because attackers can steal data, hijack accounts, or crash systems.

6. Daily Applications & Attack Surface Mapping

Application	Possible Attack Surface
Email	Phishing, malware attachments
WhatsApp	Fake links, account takeover
Banking App	Credential theft, insecure networks
Social Media	Password reuse, fake logins

7. Data Flow Explanation

User → Application → Server → Database

Example:

User enters login details

Application sends data to server

Server validates data

Database stores or retrieves information

8. Where Attacks Can Happen

During data input (phishing, fake forms)

During data transfer (Man-in-the-Middle attack)

On the server (misconfigurations)

In the database (SQL injection)

9. Summary

Cyber security is essential to protect digital data and systems. Understanding attackers, attack surfaces, and vulnerabilities like those in OWASP Top 10 helps in building secure applications. This task helped build a strong foundation in cyber security concepts.

10. Pseudo Code: Application Data Flow and Attack Points

Stage	Possible Attack
User Input	Phishing
Data Transfer	MITM
Server	Misconfiguration
Database	SQL Injection

User → Application → Server → Database

↑ ↓
Phishing SQL Injection

11. Conclusion :

This task strengthened my understanding of fundamental cyber security principles, including the CIA Triad, attack surfaces, and OWASP Top 10 vulnerabilities. It enhanced my ability to identify security risks in application data flow and prepares me for deeper, hands-on cyber security learning in the future.